EV Market Insights Research

Reference Taxonomy Note

6-Stage Consumer Funnel and 8-Family Topic Umbrella Design

Purpose
A clean reference taxonomy for topic refinement, topic-family assignment, market-funnel analysis, and later paper writing across all subreddits, submissions, and comments.

This framework deliberately separates **where the consumer is in the journey** from **what the discussion is about**. In practice, each topic can receive both a funnel stage and a topic family. That separation makes the taxonomy flexible, interpretable, and reusable.

How to use this note

Use the **6-stage funnel** in the market-funnel analyzer to capture journey position. Use the **8-family taxonomy** in the refine pipeline to group semantically related topics, including make/model-specific discussions. Only near-duplicates should be merged; broader themes should remain separate but linked through the family label.

I. Universal 6-Stage EV Consumer Funnel

The funnel is designed to be broad enough for universal project use, but EV-specific enough to capture the practical realities of charging, incentives, reliability, and post-purchase adaptation.

Stage	Definition	Typical topic signals
1. Awareness / Need Formation	Earliest discovery and interest stage. Users are still asking whether an EV is worth considering and what problem or lifestyle need it might solve.	general EV curiosity; first EV questions; should I buy an EV; EV vs regular car at a high level
2. Consideration / Shortlisting	Users begin narrowing options and comparing brands, models, classes, or broad alternatives.	Tesla vs Ioniq 5; best EV SUV; used Bolt vs Leaf; hybrid vs EV; shortlist discussions
3. Evaluation / Practical Fit	Users test whether an EV works for their life economically and operationally.	charging feasibility; apartment charging; range fit; winter performance; insurance; incentives; TCO

Stage	Definition	Typical topic signals
4. Purchase / Transaction	The discussion shifts from deciding to executing the purchase.	dealer friction; markup; financing; lease terms; order process; inventory; delivery timing
5. Onboarding / Early Ownership	Immediate post-purchase adaptation stage, where users learn the product and the ecosystem.	first charging setup; app confusion; learning features; initial service questions; first-week impressions
6. Long-Term Ownership / Retention	Later-stage ownership experience, including satisfaction, reliability, maintenance, and replacement decisions.	battery degradation; maintenance; warranty; service centers; resale; long-term satisfaction or regret

Recommended interpretation rule

Assign the funnel stage based on the **consumer's position in the journey**, not only on the topic keywords. For example, charging can appear in both Evaluation / Practical Fit and Onboarding / Early Ownership depending on whether the user is still deciding or has already purchased the vehicle.

II. Universal 8-Family Topic Umbrella Taxonomy

Topic families capture the **theme of discussion** rather than the consumer's stage. This taxonomy is intended for the refine pipeline and should be stable across subreddits, brands, and future extensions of the project.

Family	Scope	Examples
1. EV Purchase and Vehicle Selection	Brand/model choice, body class, fit-for-use, shortlist logic, and new vs used EV selection.	best EV for commuting; Model Y vs Ioniq 5; used EV recommendation
2. Charging and Energy Access	Home charging, public charging, apartment access, charging speed, compatibility, and charging convenience.	Level 2 setup; charger availability; apartment charging challenge
3. Battery, Range, and Real-World Performance	Battery health, degradation, range anxiety, road-trip performance, winter range, and efficiency under real conditions.	winter range drop; battery degradation; highway efficiency
4. Cost, Incentives, and Financial Feasibility	Price, monthly payment, insurance, tax credits, rebates, financing, leasing, and total cost of ownership.	federal tax credit; lease deals; insurance cost; TCO
5. Dealer, Purchase Process, and Market Access	Dealer interactions, order flow, wait times, markup, inventory access, and purchase transaction friction.	dealer markup; inventory shortage; delivery timing
6. Reliability, Service, and Ownership Experience	Maintenance, repair, service center quality, warranty, software issues after purchase, and overall ownership sentiment.	service delays; warranty coverage; repair frustration
7. Policy, Regulation, and Market Environment	Government policy, eligibility rules, emissions regulation, infrastructure policy, and broad market environment discourse.	tax-credit rule changes; charging policy; state regulations
8. ICE / Hybrid / EV Transition Narratives	Consumer transition stories, skepticism, education, habit shifts, and comparisons across propulsion types.	switching from ICE to EV; hybrid vs EV; conversion concerns

Recommended interpretation rule

Assign the topic family based on the **dominant subject matter**. Make/model-specific topics should usually roll up into one of these families rather than becoming new families. For example, many vehicle-class or make-specific comparison topics will still belong to **EV Purchase and Vehicle Selection**.

III. How the Two Taxonomies Work Together

Each topic can receive both a funnel stage and a family label. This dual assignment improves interpretability and helps distinguish consumer journey position from substantive topic content.

Example topic	Funnel stage	Topic family
Tesla Model Y vs Hyundai Ioniq 5 for a family car	Consideration / Shortlisting	EV Purchase and Vehicle Selection
Can I make an EV work if I live in an apartment and street park?	Evaluation / Practical Fit	Charging and Energy Access
I just bought a Bolt - how should I set up home charging and the app?	Onboarding / Early Ownership	Charging and Energy Access
My EV insurance is much higher than expected - is this normal?	Evaluation / Practical Fit	Cost, Incentives, and Financial Feasibility
Dealer added a markup and the delivery keeps slipping	Purchase / Transaction	Dealer, Purchase Process, and Market Access
After two winters, my battery range seems noticeably worse	Long-Term Ownership / Retention	Battery, Range, and Real-World Performance

Implementation notes

- Use the 6-stage taxonomy in **market_funnel_analyzer.py** as the default stage system.

- Use the 8-family taxonomy in the **topic_refine_pipeline.py** as the default umbrella system.

- Keep **merge groups** conservative: merge only near-duplicates. Use **topic families** for broader semantic grouping.

- Preserve the original BERTopic topic ID and add new fields such as **merge_group_id**, **merge_parent_topic**, **topic_family_id**, and **topic_family_label**.

Bottom line
For this research project, the recommended universal design is a **6-stage EV funnel** plus an **8-family topic umbrella taxonomy**. The funnel tells you where the consumer is; the family tells you what the conversation is about.